

RESEARCH PROJECT REPORT

Numerical Modelling and Design of Edge-Stiffened Perforated Cold-Formed High-Strength Steel Channel Columns

Conference Research Project

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Research Project Portfolio

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Abstract

Cold-formed steel (CFS) channel sections are increasingly utilized as load-bearing components in modern building structures due to their structural efficiency and ease of fabrication. These sections often incorporate web perforations to facilitate the integration of services such as electrical and signal wiring, plumbing, air-conditioning systems, structural optimization, and maintenance access. Conventionally, these perforations are created by punching and are left unstiffened, which limits their size and spacing due to the resulting reductions in structural performance. Recently, researchers have addressed this issue by introducing edge-stiffening around the perforations, and interest in this approach is growing. Concurrently, significant advancements in steel manufacturing technologies have enabled the mass production of high-strength steel (HSS) with nominal yield strengths exceeding 460 MPa. The superior mechanical properties of HSS, including increased yield strength and tensile capacity, make it highly suitable for long-span and high-rise structures, enabling reduced cross-sectional dimensions and improved connection performance. When edge-stiffening is combined with the use of HSS and optimized metal forming techniques, the structural efficiency of perforated CFS channel sections can be significantly enhanced. Despite these advantages, limited research has been conducted on the compressive behavior of edge-stiffened, perforated HSS channel sections. To address this gap, the present study undertakes a comprehensive investigation using validated numerical models to assess their axial load performance. The findings indicate that current international design standards do not fully capture the behavior of these advanced sections, highlighting the need for revised and improved design provisions.

1 Introduction

In recent years, cold-formed steel (CFS) channel sections have been increasingly adopted in building construction due to their high strength-to-weight ratio and ease of fabrication [1, 2]. Such sections frequently require web openings to accommodate building services (e.g., plumbing, electrical conduits), but these perforations introduce stress concentrations that significantly reduce buckling and compressive capacity [1, 3]. Early studies by TG. Singh and KD. Singh [4] on SHS/RHS stub columns and D. Moen and Schafer [1] on lipped channels with slotted holes demonstrated that even small openings can sharply reduce post-peak ductility and alter buckling modes. Yu [3] investigated CFS C-section members with edge-stiffened circular web holes and demonstrated that edge stiffening can substantially improve the flexural performance of members containing web openings. Subsequently, Yao [8] examined slender lipped channels with rectangular and circular holes. The distortional buckling behaviour of perforated beams was investigated by Yuan et al.

[9] and Nan-ting Yu et al. [10], who proposed analytical solutions based on Hancock's modified formulation.

Parallel parametric studies and experiments addressed axial compression under various loading conditions. Chen et al. [5] performed an extensive FE parametric study on edge-stiffened circular holes to propose simplified design equations. Laboratory tests on full-length channels with edge-stiffened and unstiffened web holes, reported by Chen et al. [6] and Roy et al. [7], confirmed that edge-stiffened holes enhance axial capacity while unstiffened holes reduce it under compression. Chen et al. [12] and Chandramohan et al. [13] reported that edge-stiffened web holes increased moment capacity by 14.5% (circular) and 9% (elongated) compared to unstiffened holes, respectively, and found existing design equations overly conservative. Chen et al. [14] reported a 15.4% moment capacity rise for back-to-back channels with edge-stiffened holes, versus a 15.1% fall for unstiffened holes. Edge-stiffened circular holes were found to greatly improve web crippling strength under one-flange and end-one-flange loading, reducing strength loss for unstiffened holes, leading to new reduction factor equations [15, 16, 17]. Parametric finite element studies on web crippling of cold-formed steel channels with edge-stiffened web holes under end-one-flange, interior-two-flange, and end-two-flange loading conditions were conducted, leading to proposed design reduction factors and direct strength method equations that account for the beneficial effect of edge stiffeners [18, 19, 20]. Fang et al. [21] examined web crippling of CFS channels with web holes at 20–700°C, found reduction factors sensitive to hole geometry but not temperature, and proposed design equations. Yousefi et al. [22] found existing web crippling rules unconservative for ferritic stainless steel channels and proposed unified equations. Xie et al. [23] found edge-stiffened holes reduced combined strength by 4.2% while unstiffened holes caused a 22.1% reduction under combined web crippling and bending. Elastic shear buckling of perforated CFS channels was investigated numerically [24, 25, 26] where edge stiffening improved shear capacity, leading to modified shear buckling coefficients and DSM-based strength assessments. Wang et al. [27] found that edge stiffeners around large web openings restore combined bending-shear capacity, with reinforcement recommended when hole depth exceeds half the web height. Chen et al. [28] tested sigma-shaped CFS sections with web holes under ITF loading, reporting a 24% average reduction in web-bearing resistance, with offset holes more detrimental than centred holes. The flexural behaviour of press-braked S690 channel beams and the web crippling response of S690/S960 unlipped channels under ITF loading were investigated, revealing inaccuracies in current design standards and leading to improved predictive approaches [29, 30]. Xue et al. [31] tested Q1200 ultra-high strength steel angle and channel stub columns, finding that existing design rules were unsafe for channels and overly conservative for slender angles, and proposed strain-hardening-based modifications. The axial compressive behaviour of LQ550 high-strength CFS columns with web holes was investigated by Wang et al. [32], revealing that web stiffeners delay local

buckling and prompting a proposed reduction coefficient. While the axial compressive behavior of standard CFS columns with edge-stiffened holes has been explored, no research has yet investigated the numerical modelling and design of perforated high-strength steel (HSS) channel columns featuring edge-stiffened web holes. Therefore, this study aims to develop a validated non-linear finite element model to evaluate the structural behavior of edge-stiffened perforated HSS columns, followed by an extensive parametric study to propose modifications to current design guidelines.

2 Research Gap

Previous investigations have examined the structural behaviour of cold-formed steel sections with web openings, including unstiffened and edge-stiffened configurations.

However, the behaviour of perforated high-strength steel channel columns with edge-stiffened openings remains comparatively less investigated.

In particular, the interaction between high-strength steel, web perforations, edge stiffening, and member slenderness requires further investigation.

The present study addresses this research gap through a systematic finite element investigation of non-perforated, unstiffened perforated, and edge-stiffened perforated high-strength steel channel columns.

3 Research Objectives

The primary objectives of the research are:

1. To develop a nonlinear finite element model of cold-formed high-strength steel channel columns using ABAQUS.
2. To investigate the influence of web perforations on the axial compressive capacity of the channel sections.
3. To compare the structural response of non-perforated, unstiffened perforated, and edge-stiffened perforated sections.
4. To investigate the influence of column length and member slenderness on axial load carrying capacity.
5. To examine the buckling behaviour of the investigated channel sections.

6. To assess the applicability of existing design approaches to the investigated structural configurations.

4 Section Geometry and Configurations

Two principal channel section configurations were investigated, designated as Section 190 and Section 240, corresponding to web depths of 190 mm and 240 mm, respectively.

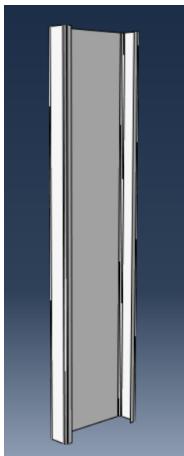
The investigated sections included:

- Non-perforated channel sections,
- Channel sections with unstiffened web openings, and
- Channel sections with edge-stiffened web openings.

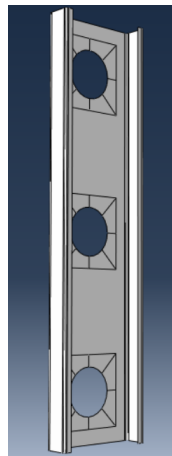
5 Perforation and Edge-Stiffening Configuration

Three principal web configurations were considered in the numerical investigation:

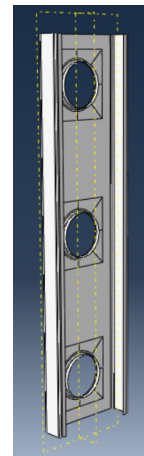
1. Plain or non-perforated web,
2. Unstiffened web opening, and
3. Edge-stiffened web opening.



(a) Plain web



(b) Unstiffened hole



(c) Edge-stiffened hole

Figure 1: Investigated web configurations.

The comparison between these three configurations forms the primary basis for assessing the influence of web perforations and edge stiffening on the axial response of the channel columns.

6 Finite Element Modelling

A three-dimensional nonlinear finite element model was developed using ABAQUS to investigate the axial compressive behaviour of the channel columns.

The modelling methodology was developed based on established numerical modelling approaches for cold-formed steel channel sections, with appropriate modifications to represent the high-strength steel material used in the present investigation.

6.1 Geometry Modelling

The channel sections were modelled by explicitly representing the web, flanges, edge lips, corner regions, and web openings.

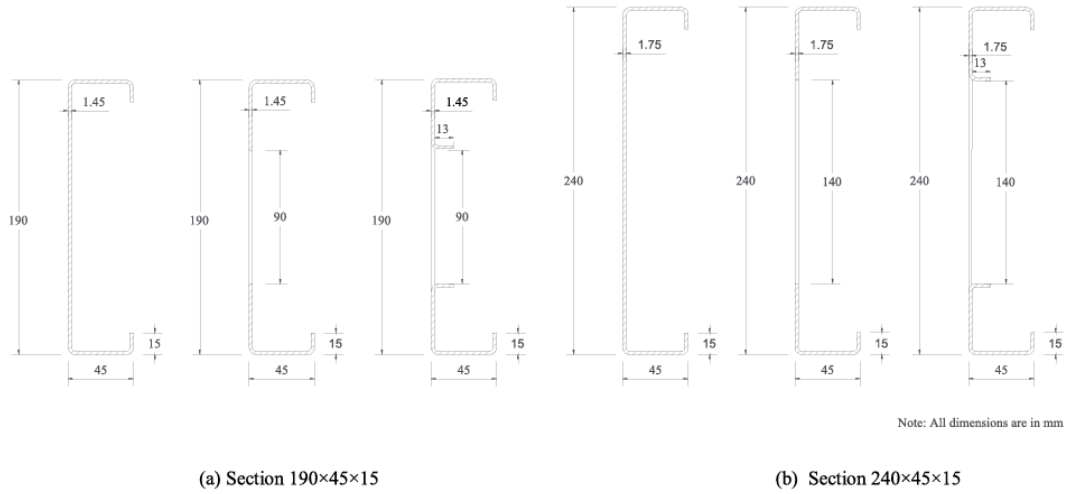


Figure 2: Three-dimensional finite element model of the channel column.

7 Material Properties

S700 high-strength steel was adopted in the finite element models.

The Young's modulus and Poisson's ratio were taken as

$$E = 212 \text{ GPa} \quad (1)$$

and

$$\nu = 0.30. \quad (2)$$

Separate material properties were assigned to the flat and corner regions to account for the strength enhancement associated with the cold-forming process.

The adopted material properties are summarised in Table 1.

Table 1: Material properties used in the FE model.

Property	Flat Region	Corner Region
Yield stress	719 MPa	897 MPa
Ultimate tensile stress	840 MPa	983 MPa
Young's modulus	212 GPa	212 GPa
Poisson's ratio	0.30	0.30

Engineering stress–strain data were converted into true stress and true plastic strain for implementation in ABAQUS.

8 Finite Element Mesh

The channel columns were discretised using four-node quadrilateral reduced-integration shell elements (S4R).

A nominal mesh size of approximately $8 \text{ mm} \times 8 \text{ mm}$ was adopted, with local refinement provided near the web perforations, edge stiffeners, and corner regions.

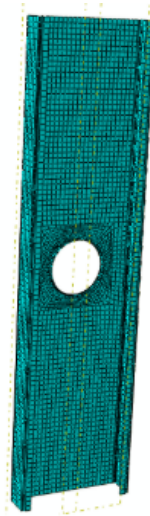


Figure 3: Finite element mesh of the channel column with local refinement around the web opening.

9 Boundary Conditions and Loading

Pin-pin end conditions were adopted for the numerical models.

Reference points were positioned beyond the physical ends of the specimens. The end cross-sections were coupled to the corresponding reference points using coupling constraints.

The axial compressive load was applied through the loading reference point and aligned with the centre of gravity of the cross-section.

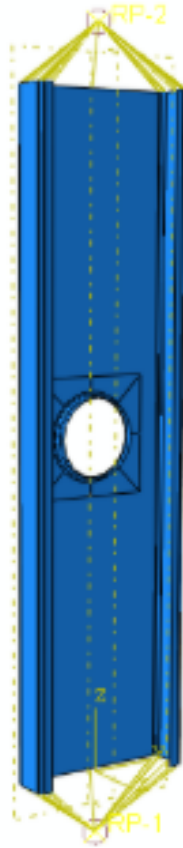


Figure 4: Boundary conditions and axial loading arrangement.

10 Initial Geometric Imperfections

Initial geometric imperfections were incorporated into the finite element models because of their influence on the buckling behaviour of thin-walled cold-formed steel members.

Elastic eigenvalue analysis was first conducted to obtain the relevant buckling modes. The selected eigenmodes were subsequently scaled to the prescribed amplitudes and introduced into the nominal geometry.

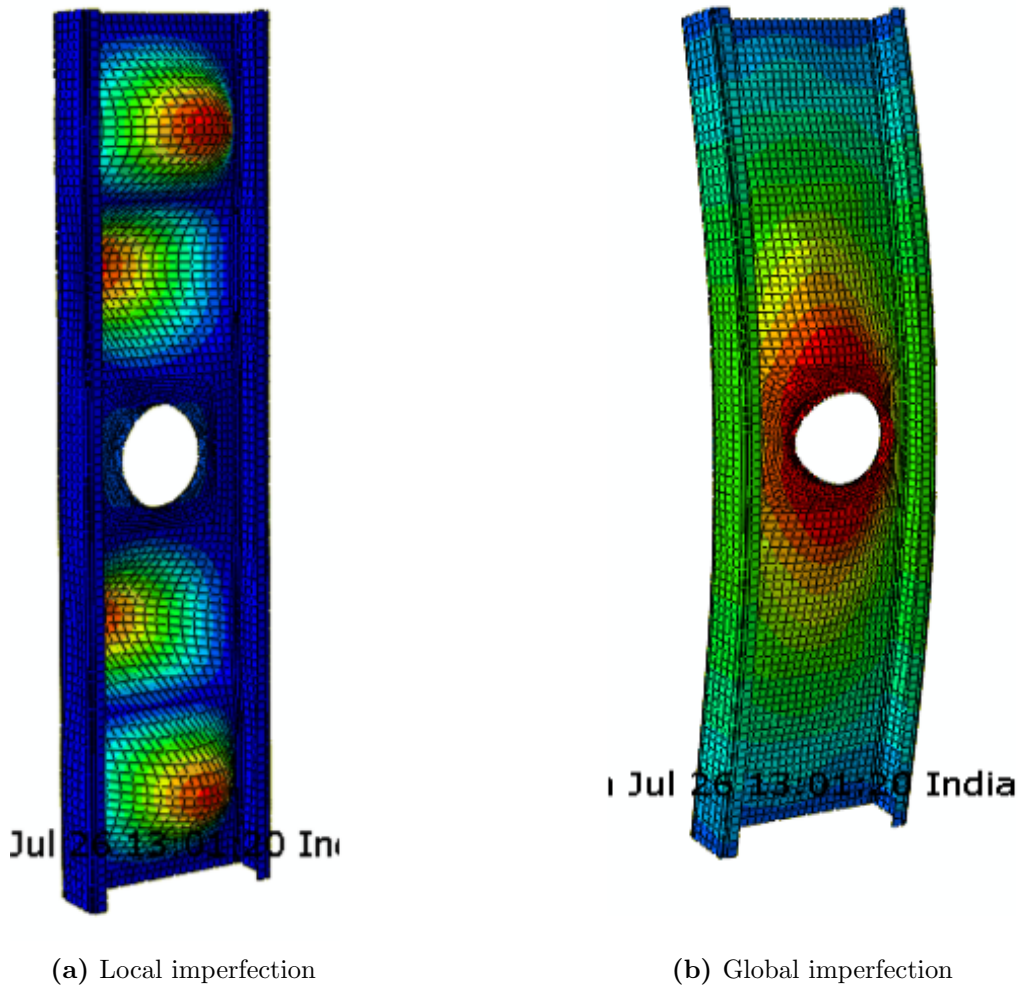


Figure 5: Representative initial geometric imperfections used in the finite element analysis.

The local imperfection amplitude was taken as 0.5% of the nominal section thickness.

The global imperfection amplitude was taken as

$$w_0 = \frac{L}{1000} \quad (3)$$

where L is the overall column length.

11 Nonlinear Analysis Procedure

The nonlinear equilibrium path was traced using the modified Riks method.

The Riks method enables the numerical response to be followed through the onset of instability and into the post-buckling regime.

Geometric nonlinearity and the prescribed initial geometric imperfections were included

throughout the nonlinear analysis.

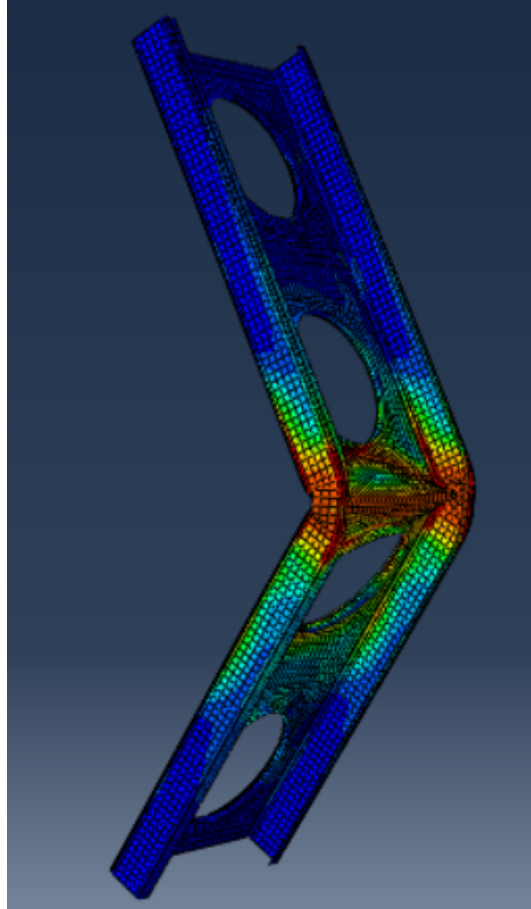


Figure 6: Representative nonlinear load–displacement response obtained from the finite element analysis.

12 Parametric Study

Following the development of the finite element modelling framework, a systematic parametric study was conducted.

A total of 348 finite element models were developed to investigate the axial response of the investigated channel sections.

The primary parameters considered in the study were:

- Section depth,
- Column length,
- Web configuration, and
- Presence or absence of edge stiffening around the web opening.

The column length was varied over the investigated range to evaluate the influence of member slenderness on axial capacity.

13 Results and Discussion

13.1 Effect of Web Configuration

The first part of the numerical investigation focused on comparing three web configurations:

1. Non-perforated sections,
2. Sections with unstiffened web holes, and
3. Sections with edge-stiffened web holes.

The comparison was performed in terms of ultimate axial capacity, load–displacement response, and observed buckling behaviour.

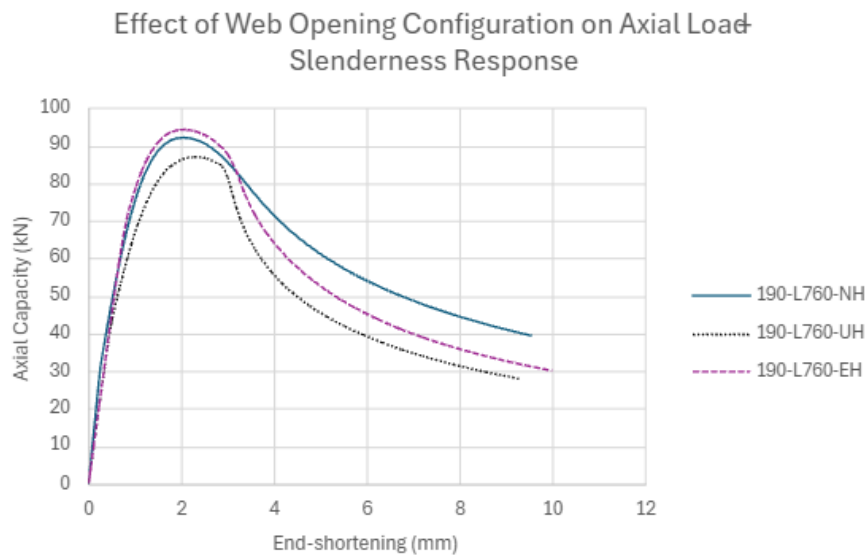


Figure 7: Comparison of axial capacities for plain, unstiffened, and edge-stiffened web configurations.

The numerical results demonstrate the influence of the presence of web openings on the axial response of the channel columns. The comparison between unstiffened and edge-stiffened openings further illustrates the effect of local stiffening around the perforation.

13.2 Effect of Column Length

Column length has a direct influence on member slenderness and therefore on the governing buckling response of the channel columns.

The numerical investigation considered column lengths ranging from 380 mm to 3360 mm.

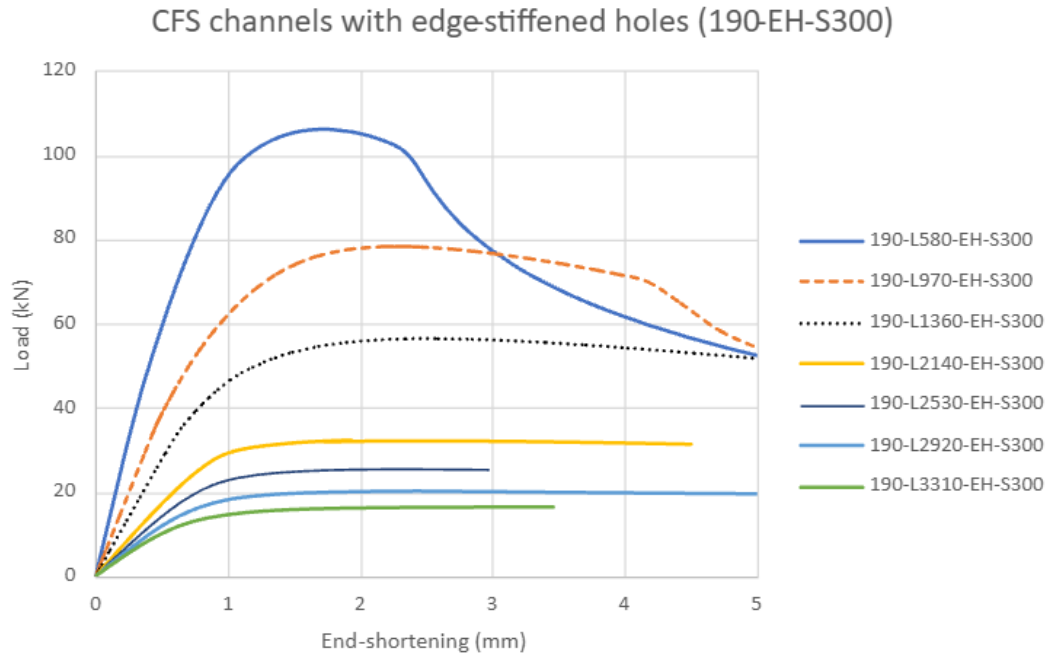


Figure 8: Variation of ultimate axial capacity with column length.

The variation in axial capacity with column length was analysed for the different web configurations.

For shorter members, local cross-sectional behaviour can have a greater influence on the response, whereas increasing member length increases the importance of overall member instability.

14 Design Assessment

The finite element predictions were compared with relevant design approaches to examine their ability to represent the axial resistance of the investigated channel columns.

The comparison focused on:

- Predicted axial capacity,
- Finite element ultimate capacity,

- Influence of web perforations,
- Influence of edge stiffening, and
- Influence of member slenderness.

15 Key Findings

The principal findings of the numerical investigation are summarised below:

1. Web perforations influence the axial load carrying capacity and buckling response of cold-formed high-strength steel channel columns.
2. The structural response of unstiffened perforated sections differs from that of equivalent non-perforated sections because of the interruption of the web and associated changes in local stiffness.
3. Edge stiffening around the web opening modifies the local structural behaviour and can improve the performance of perforated members.
4. Increasing column length increases member slenderness and changes the governing instability behaviour.
5. The nonlinear finite element modelling framework provides a systematic means of investigating the interaction between web perforations, edge stiffening, and member slenderness.
6. The results from the parametric study provide a numerical database for further assessment of design provisions for perforated high-strength cold-formed steel channel columns.

16 Research Contribution

My contribution to this research project included:

- Development and execution of finite element models in ABAQUS.
- Modelling of cold-formed high-strength steel channel sections with different web configurations.
- Implementation of non-perforated, unstiffened perforated, and edge-stiffened perforated configurations.

- Implementation and analysis of initial geometric imperfections.
- Execution of nonlinear Riks analyses.
- Development and execution of a large parametric finite element study.
- Post-processing and interpretation of numerical results.
- Investigation of the influence of column length and web configuration on axial capacity.
- Preparation and interpretation of figures and numerical results for the research work.
- Contribution to the research presented at ICCMS 2025.

17 Conference Presentation

The research was presented at the:

**10th International Congress on Computational Mechanics and Simulations
(ICCMS 2025)**

Indian Institute of Technology Bhubaneswar
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Presented Research

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18 Software and Computational Tools

The research utilised the following computational tools:

- ABAQUS – finite element modelling and nonlinear analysis.

- Python – numerical processing and automation.
- CUFSM / finite strip analysis – elastic buckling analysis, where applicable.
- Microsoft Excel – data processing and engineering calculations.

19 Conclusion

A nonlinear finite element modelling framework was developed to investigate the axial compressive behaviour of edge-stiffened perforated cold-formed high-strength steel channel columns.

The investigation considered non-perforated, unstiffened perforated, and edge-stiffened perforated configurations. A systematic parametric study involving 348 finite element models was conducted, with particular attention given to the effects of web configuration and column length.

The numerical investigation provides insight into the structural behaviour of perforated high-strength cold-formed steel channel columns and establishes a computational basis for further research into their design and structural optimisation.

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